

Telemedicine Access for Rural Healthcare in Nabha

Team-17

UML Design Documentation

Project Overview

Access to healthcare in rural regions is often limited due to lack of infrastructure and specialists. To address this issue, this project introduces a mobile-based telemedicine platform that enables remote consultations, digital record management, and real-time pharmacy integration in rural Nabha.

System Specifications

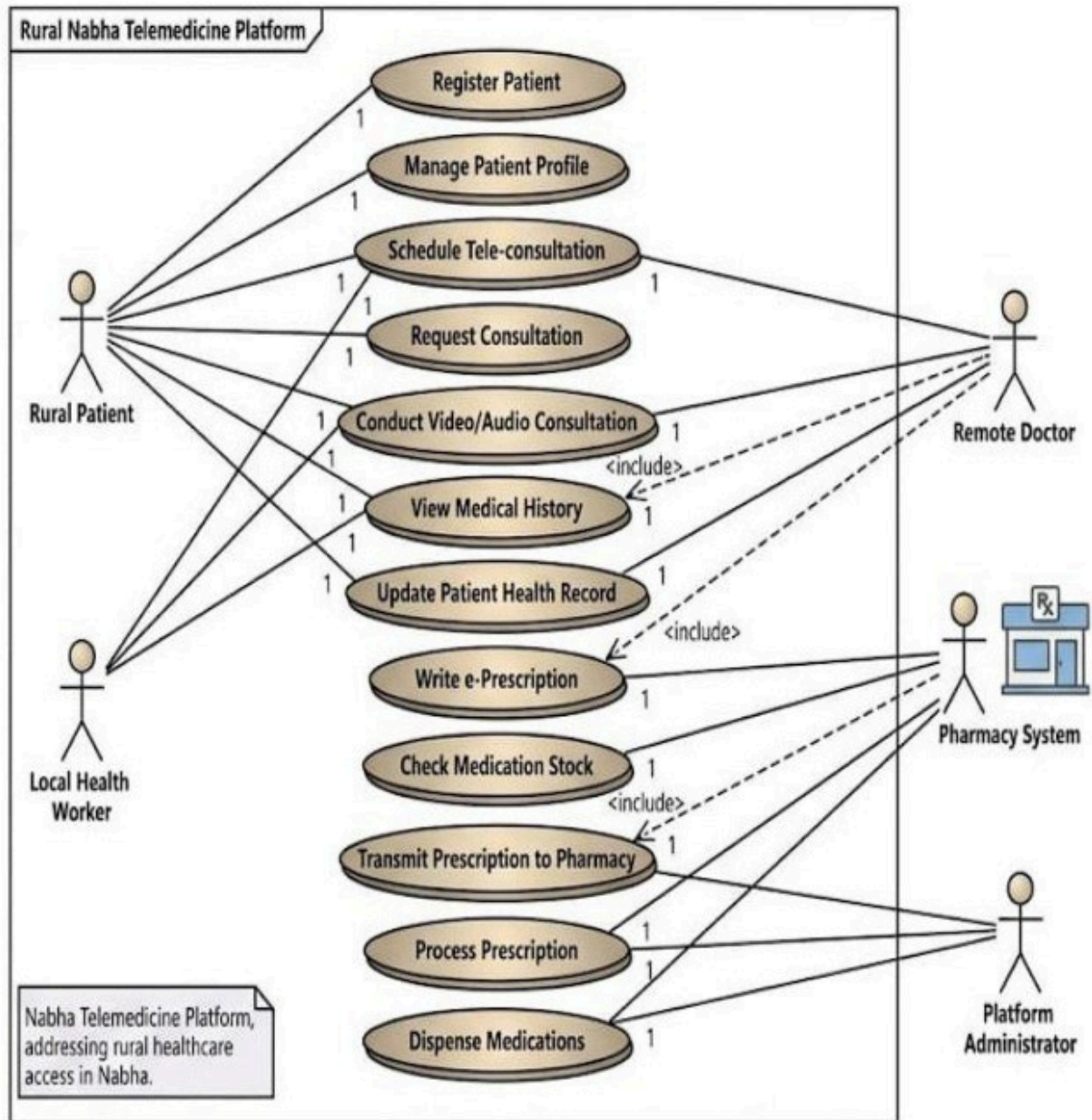
Frontend	Flutter(Mobile App)
Backend	Node.js+Express.js
Database	MongoDB
Video Service	WebRTC
Authentication	Login+JWT
Notifications	Push Notifications
API Communication	Restful APIs
Security	Role Based Access Control
Operating Platform	Android,ioS

Diagrams in this Document

	Diagram	Scope
1	Use Case Diagram	Shows users and system functions
2	Class Diagram	Shows system classes and relationships
3	Sequence Diagram	Shows interaction flow between components
4	State Diagram	Shows states of an appointment
5	Activity Diagram — Login & Booking	Shows login and booking process
6	Activity Diagram — Consultation	Shows consultation workflow
7	Activity Diagram — Analytics & Reports	Analytics access and PDF report export
8	Component Diagram	Shows system components
9	Deployment Diagram	Shows system deployment setup

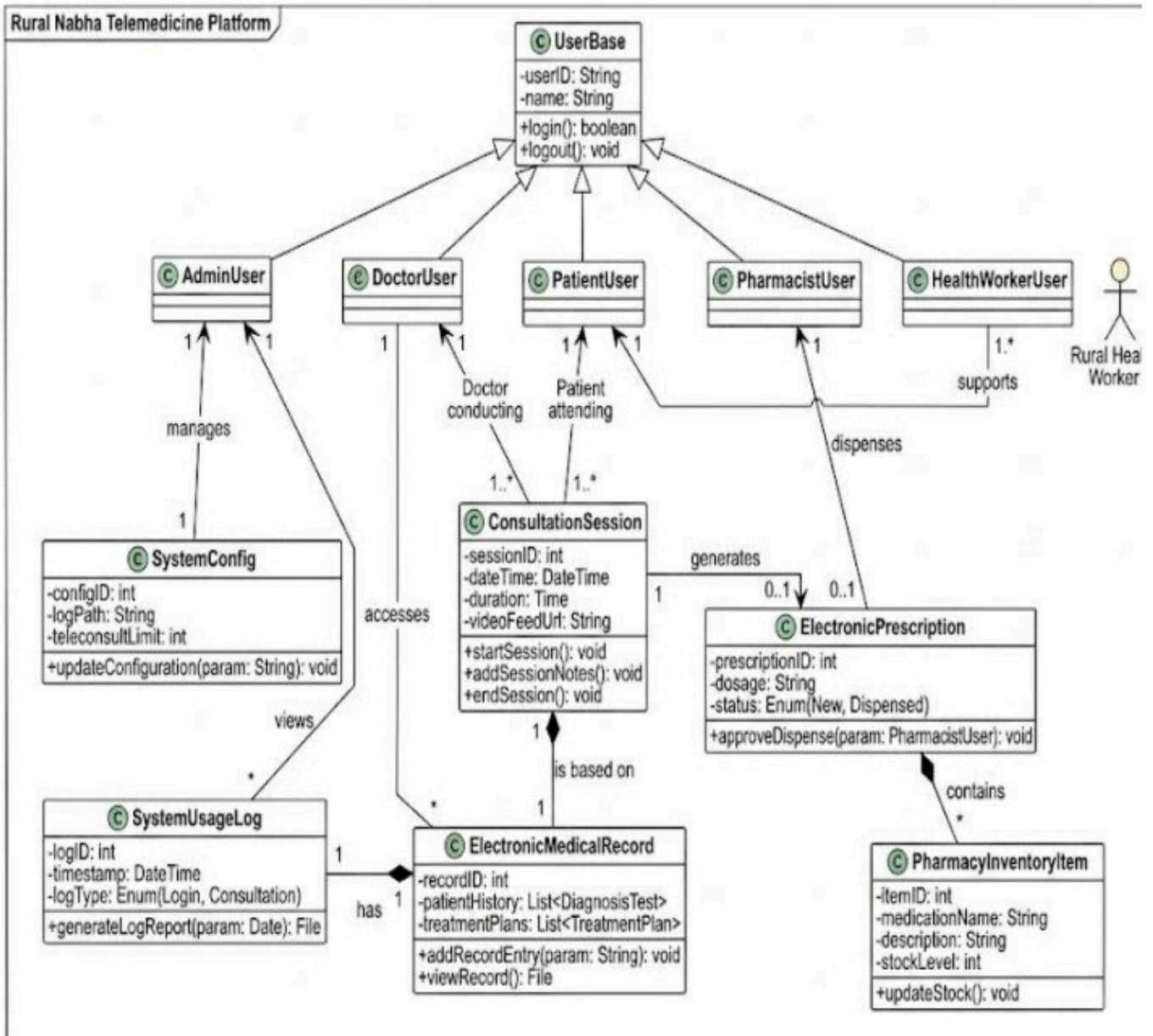
USE CASE DIAGRAM

Maps actors (Admin, Operator, MySQL DB, ML Engine, SHAP Explainer) to system use cases including schedule management, maintenance workflows, ML predictions, report generation, and RBAC login.



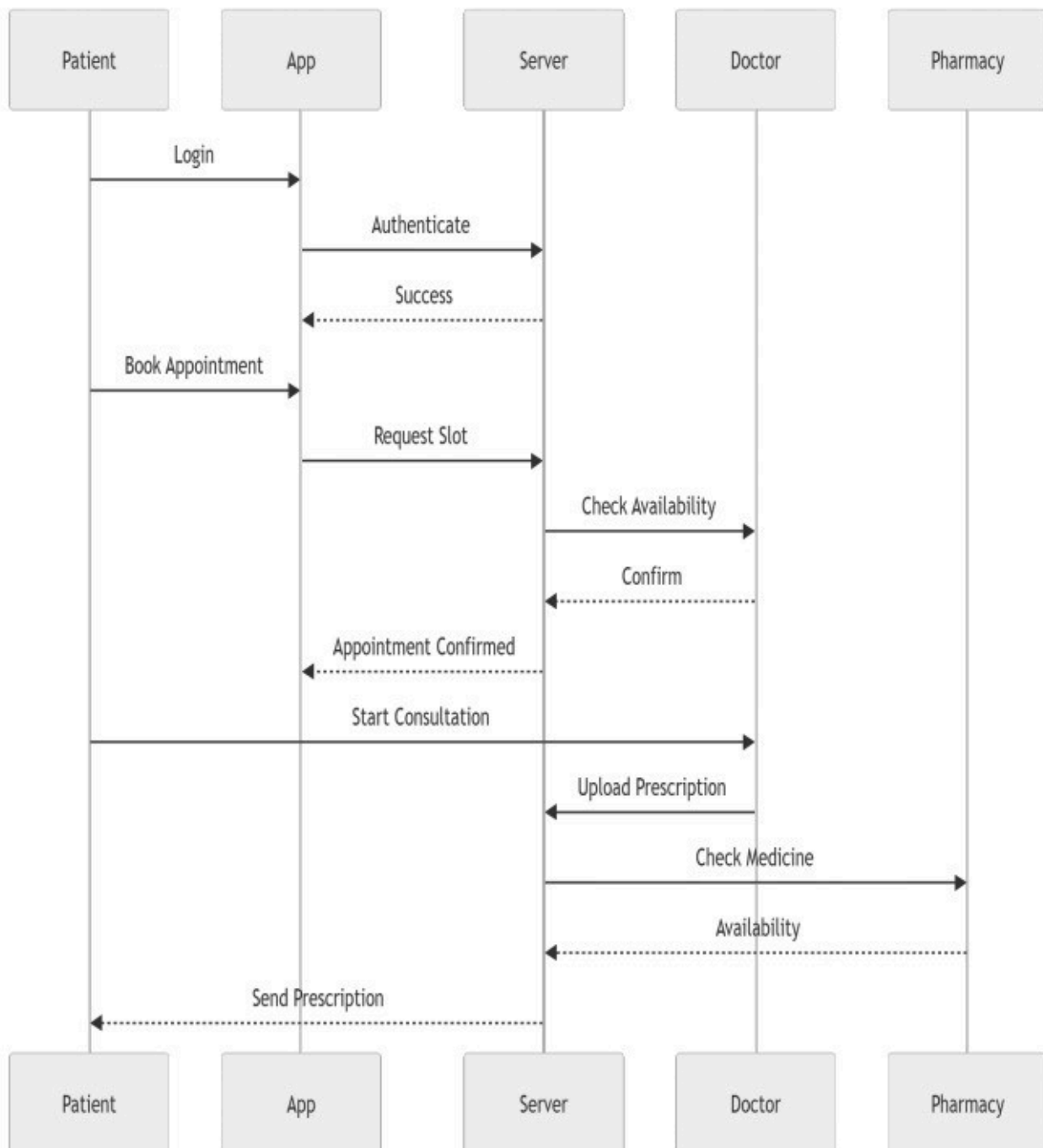
CLASS DIAGRAM

Defines the static structure of TransitFlow — core classes (User, Train, Schedule, Route, MLModel, HealthPrediction, MaintenanceJobCard, Alert, Report, SystemConfig), their attributes, methods, and relationships.



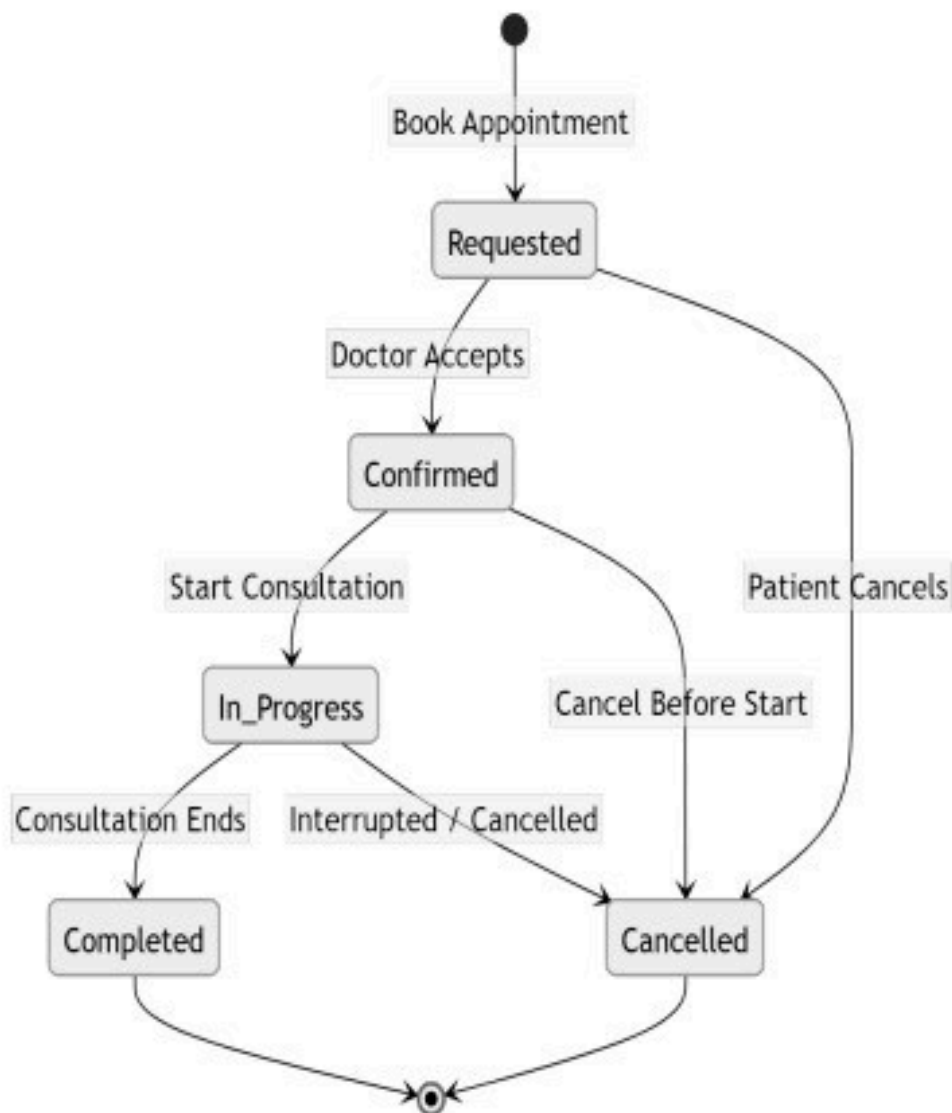
SEQUENCE DIAGRAM

Time-ordered message flow across four flows: Authentication (RBAC), Fleet KPI Dashboard loading, ML Health Prediction (XGBoost + SHAP), and Schedule Generation (Gantt Chart).



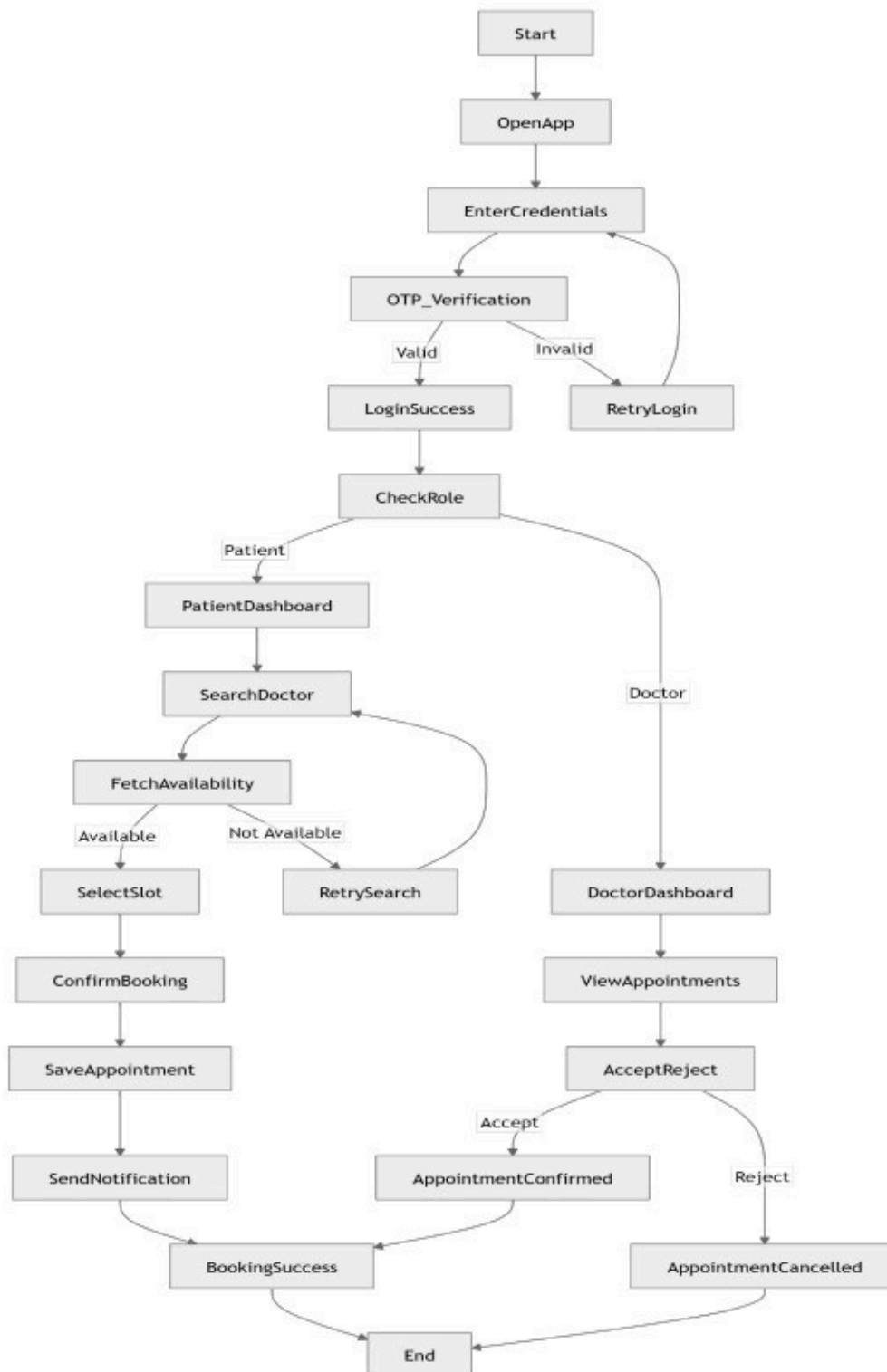
STATE DIAGRAM

Models the train lifecycle: Idle → InService → Faulty / MaintenanceScheduled → UnderMaintenance → Decommissioned, driven by ML alerts, schedule events, and repair outcomes.



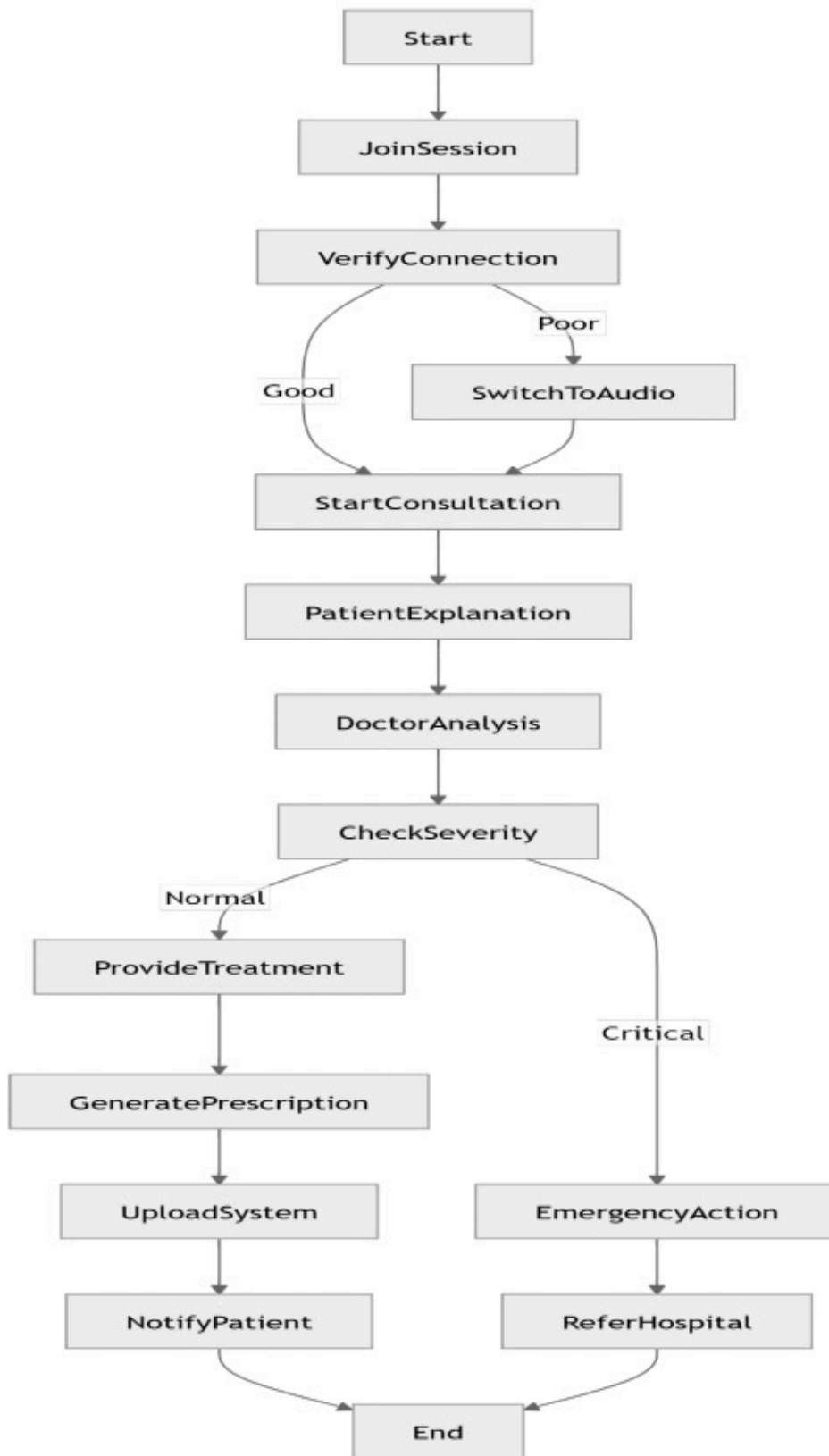
ACTIVITY DIAGRAM — Login & Booking

Flow from app launch through RBAC login, role-based access branching (Admin / Operator), real-time alert check, and emergency job card creation or daily planning continuation.



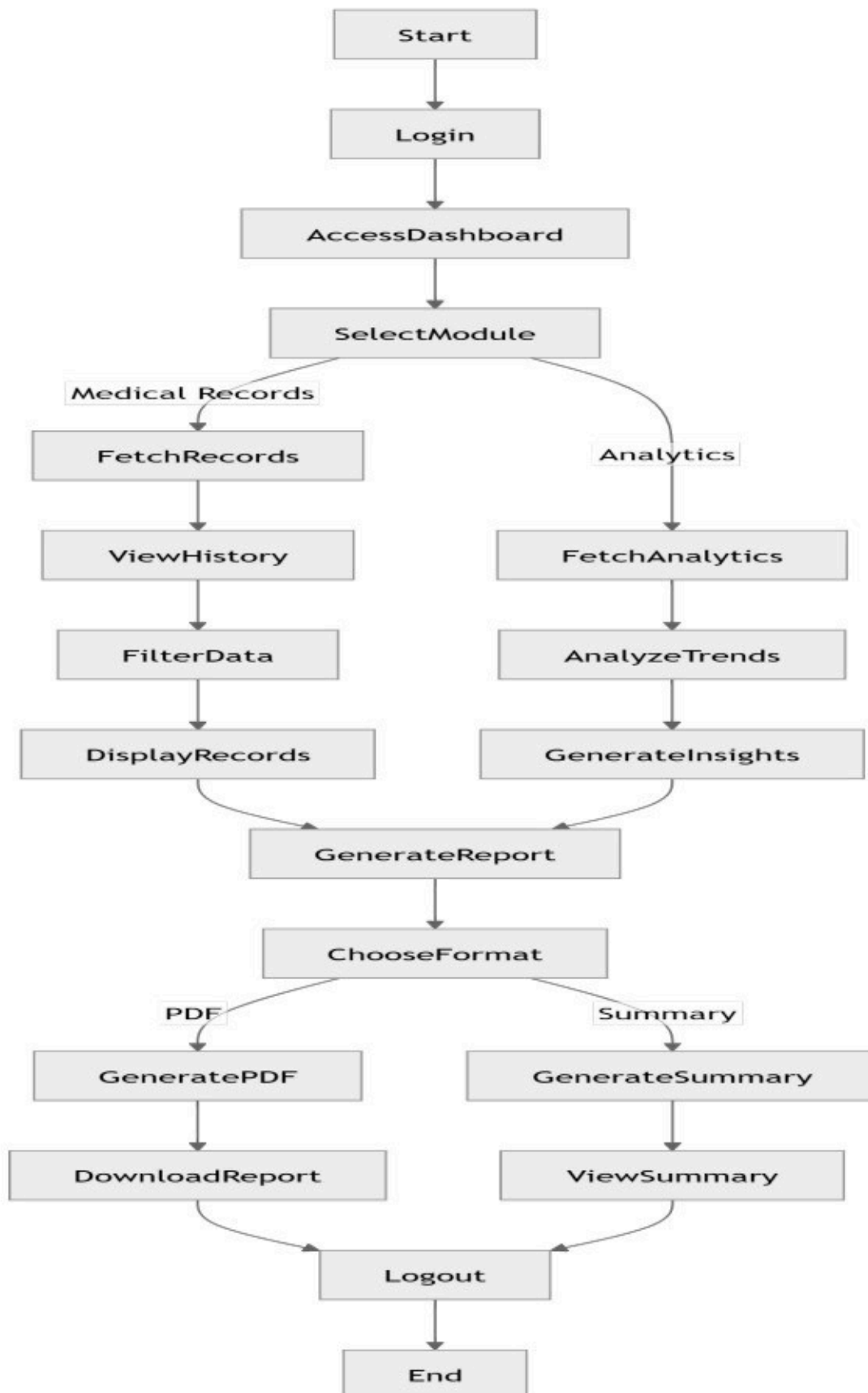
ACTIVITY DIAGRAM — Consultation

Parallel flows: Scheduling Module (fetch trains/routes → assign → save to MySQL → Gantt Chart) and ML Insights (fetch features → XGBoost → SHAP → display health status). Branches on predicted risk into Maintenance Module or Analytics.



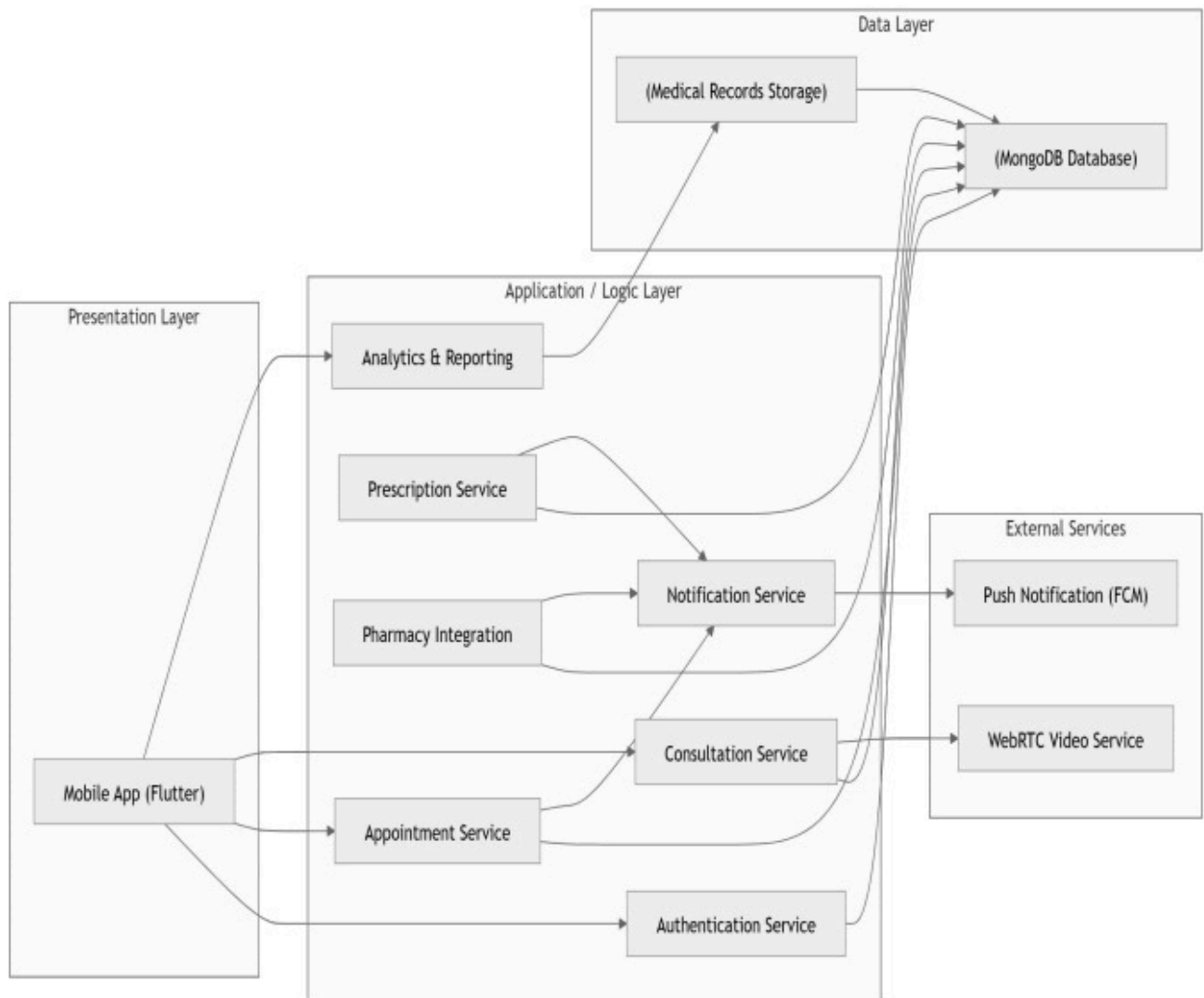
ACTIVITY DIAGRAM — Analytics & Reporting

Analytics Module access, historical fleet trend review, optional multi-tab PDF report generation (type selection → generate → export), and session logout.



COMPONENT DIAGRAM

Three-layer architecture: Presentation (9 Streamlit UI pages), Application/Logic (Auth, Alert Engine, ML Inference, Scheduling, Maintenance, Report, Config), and Data Layer (MySQL TransitFlow DB, 7 tables) with ML Pipeline and Output/Export subsystems.



DEPLOYMENT DIAGRAM

Physical nodes: Admin Workstation (Streamlit browser via HTTPS:8501), HMRL Application Server (Python/Linux VM), HMRL ML Server (GPU VM — XGBoost, SHAP, Model Registry), HMRL DB Server (MySQL 8.x), and Train Fleet IoT Gateway (MQTT/REST health telemetry).

